

PHYSICS IX

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF PHYSICS.

COMPILED BY M. Zubair Arif Yousufi

Section "B" (Short Answer Question)

Section "B" (Short Answer)		YEARS
QUESTIONS		
Chapter 1: PHYSICAL QUANTITIES AND MEASUREMENT		
Section B		
5. Define density. Also write its formula and SI unit.		2025
2. Define fundamental and derived quantities. Also write three examples of each.		2024
2. Define density with formula and unit also write ways of measurement of density.		2023
2. Define: (i) Atomic Physics (ii) Nuclear Physics (iii) Plasma Physics		2022
3. Define Physics and write its two importance in daily life.		2021
2. Write down two contributions of each scientist: (i) Yaqoob Al-Kindi (ii) Ibn-Al-Haitham		2019
10. The radius of a Hydrogen atom is $0.53 \times 10^{-10} \text{m}$. Convert it in kilometer, millimeter, micrometer and nanometer.		2018
2. Define Physics and write the names of any of its four branches.		2017
2. Name and define branches of physicals about solid martial, nucleus and atoms and astronomical bodies.		2015
3. Write down the S.I. units of the following physical quantities. (i) Torque (ii) Weight (iii) Power (iv) Stress (v) Length (vi) Frequency		
SOME IMPORTANT QUESTIONS FROM NEW BOOK		
Q. 1 Write down the S.I. Units and types of the physical Quantities (i) Ampere (ii) Volume (iii) Time (iv) Temperature		

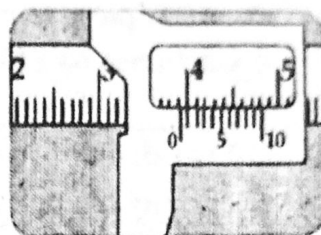
(v) Force (vi) Density (vii) Velocity (viii) Acceleration

Q. 2 Convert the following values.

- a) $230 \text{ cm} = \underline{\hspace{2cm}} \text{ m}$ b) $250 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$
 c) $0.5 \text{ s} = \underline{\hspace{2cm}} \text{ ms}$ d) $0.8 \text{ m} = \underline{\hspace{2cm}} \text{ mm}$
 e) $350 \text{ ms} = \underline{\hspace{2cm}} \text{ s}$ f) $1.2 \text{ Kg} = \underline{\hspace{2cm}} \text{ g}$

Q. 3 An engineer measures the width of an aluminum sheet using Vernier caliper as shown in figure.

- a) What is the measurement of the width of aluminum sheet?
 b) Which gives more precise measurement Vernier caliper, Screw Gauge or meter rule?

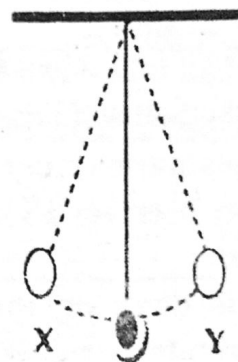


Q. 4 A pendulum swings as shown in figure 1.30 from X to Y and back to X again

i) What would be the most accurate way of measuring time for one oscillation? with the help of a Stop Watch.

- a) Record time for 10 oscillations and multiply by 10
 b) Record time for 10 oscillation and divide by 10
 c) Record time for one oscillation
 d) Record time from X to Y and double it

ii) Suggest an instrument for measuring time period more accurately.



Q. 5 Write the correct prefix of notion

- a) $75000 \text{ m} = 750 \underline{\hspace{1cm}}$ b) $2/1000 \text{ sec} = 1 \underline{\hspace{1cm}}$
 c) $1/1000000 \text{ g} = 1 \underline{\hspace{1cm}}$ d) $1000000000 \text{ m} = 1 \underline{\hspace{1cm}}$

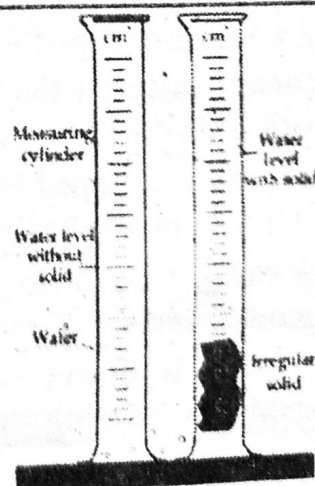
Q. 6 Write values in standard and scientific notation

- a) The radius of 1st orbit of Hydrogen atom is $r = 0.53 \text{ \AA} = \underline{\hspace{2cm}}$
 b) 1 light year is $2628000000000 \text{ m} = \underline{\hspace{2cm}}$
 c) Vacuum pressure $2.7 \times 10^{-4} \text{ torr} = \underline{\hspace{2cm}}$

Q. 7. A measuring cylinder in a given figure is filled with 500cc water. A stone of mass 20g is immersed in to the cylinder such that, water level rises up to 800cc. Which statement is correct?

- a) The difference between the readings gives the density of stone.
 b) The difference between the readings gives volume of the stone

The final reading gives the density of stone



d) The final reading gives the volume of stone

Q. 8 Write significant numbers in the following values.

a) 980 has _____ Significant numbers.

b) 91.60 has _____ Significant numbers.

c) 10010.100 has _____ Significant numbers.

d) 0.0086 has _____ Significant numbers.

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q. 1. A wooden piece is made in different shapes take length (l) = radius (r) = 2m Calculate its volume as a:

a) Sphere b) Cube c) Cylinder d) Pyramid

Q. 2. Find the density of wood as sphere and cube if the mass of wood is 1kg. Is there any change in density due to shape?

Chapter 2:

KINEMATICS

Section B

7. Write any three differences between scalar quantity and vector quantity.	2025
11. A ball falls down from top of height 90 meters. How much time the ball will take to reach the ground?	
8. A bus starts from rest and travels along a straight path its velocity becomes 15 m s^{-1} in 5 seconds. Calculate acceleration of the bus.	2024
3. Write down any three difference between Translatory motion and vibratory motion.	2023
13. A bus is moving on a road with 15 ms^{-1} and its accelerates at 5 ms^{-1} and its accelerates at 5 ms^{-2} . Find the final velocity of the bus after 6 seconds.	
8. A ball is dropped from the top of the height of 70m. How much time, the ball will take to strike the ground? ($g = 10 \text{ ms}^{-2}$)	2022
8. A car starts from rest and after 20 seconds its velocity becomes 108 km/h. Find the acceleration of the car.	2021
15. A ball is dropped from a tower it reaches the ground in 10 seconds. Calculate the height of the tower and the velocity with which it hits the ground.	2019
11. A ball is dropped from a height of 122.5m. How much time will it take to reach the ground?	2018
5. A car starting from rest attains a velocity of 20 ms^{-1} in 5 seconds. Find the distance covered by the car.	2017
3. A ball is dropped from a tower, it reaches the ground in 10 seconds Calculate the height of the tower. (Take $g=10 \text{ m/s}^2$)	2016

4. A car is moving with uniform acceleration and attains the velocity of 108 kmh^{-1} in 5 minutes. Find acceleration of the car.	2015
Section C	
16. What do you mean by relative motion? Derive second equation of motion.	2025
14. What is Kinematics? Derive the equation $2aS = v_f^2 - v_i^2$	2024
14. Explain term " V_i " and derive the equation $S = vit + \frac{1}{2} at^2$	2023
14. Define uniform velocity and derive the equation $S = v_i t + \frac{1}{2} at^2$	2022
12. What do the symbol " V_i " and " V_f " represent? And also derive the equation $V_f^2 = V_i^2 + 2aS$	2021
23. (a) Derive the equation: $S = V_i t + \frac{1}{2} at^2$	2017
24. (a) Derive the equation $2aS = V_f^2 - V_i^2$	2016
25. (a) Derive second equation of motion in five Steps.	2015
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Q.1 a) Define rest and motion.	
Q.1 b) What is meant by relative motion.	
Q.2 a) Define speed and velocity	
Q.2 b) What is difference between distance and displacement.	
Q.2 c) Define acceleration	
Q.3 a) Define scalar and vector quantities?	
Q.3 b) How represent vector quantities are represented graphically?	
Q.4 a) Define motion under gravity?	
Q.4 b) Why gravity is taken negative for an object moving in upward direction?	
SOME IMPORTANT NUMERICAL FROM NEW BOOK	
Q.1 Calculate the acceleration of a bus that speed up from 20 ms^{-1} to 40 ms^{-1} in 8 seconds.	
Q.2 A bus is moving on a road with 15 ms^{-1} and it accelerates at 5 ms^{-2} . Find the final velocity of bus after 6 seconds.	
Q.3 A car starts moving from rest with an acceleration of 5 ms^{-1} . Find out the time to travel 50m distance.	
Q.4 A ball is dropped from a height of 50m. What will be its velocity before touching ground?	
Q.5 If a body is thrown up ward with vertical velocity 50 ms^{-1} . Calculate	

maximum height which body can reach.

Q.6 A ball falls down from top of height of 70m. How much time the ball will take to reach the ground.

Chapter 3:

DYNAMICS

Section B

8. A gun with a mass of 6kg fires a bullet of mass 40gram, Its velocity is 100 m/s Calculate the recoil velocity of gun. **2025**

3. State Newton's Second Law of Motion and derive its equation. **2024**

9. A car is running on a circular path of highway having about 100 m radius. The mass of the car is 600kg and its velocity is 72 kmh^{-1} . Find centripetal force exerted by the car.

4. What is momentum? Write its formula and S.I Unit. **2023**
11. A force of 3400N is applied on a body of mass is 85000g. Find the acceleration produced by the force.

3. State Newton's Second Law of Motion and derive the equation $F=ma$ **2022**

5. Write Newton's three laws of Motion. **2021**
9. Determine the acceleration of a car of mass 900 kg, when a net force of 2700 N acts on it.
11. At what speed must an object travel in a circle of radius 3 meter to experience a centripetal acceleration of 12 m s^{-2}

5. State Newton's Second Law of Motion and Derive $F = ma$. **2019**

4. State Newton's 2nd Law of Motion and derive $F=ma$. **2018**

4. State Newton's laws of motion. **2017**
11. A car is moving in a circular track of radius 30 m at a constant speed of 20m/s. Find the centripetal acceleration of the car.

5. Write the three points of comparison between weight and mass. **2015**

9. A String 2, long is used to whirl a 100gm stone in horizontal circle at a speed of 2 ms^{-1} find tension in string.

Section C

15. Define momentum and write its S.I unit. State and prove law of conservation of momentum. **2025**

19. Write any two advantages and two disadvantages of friction. Also write two methods of reducing friction. **2024**

25. (c) Write down two advantages and two disadvantages of Frictions **2017**

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Q.1 Define momentum with SI unit?

Q.2 When a free falling object moves towards earth due to pull of earth on it. Does earth also move towards that object due to reaction? Explain.

Q.3 a) Why a wire fence is designed in the helmet of batsman?

Q.3 b) How does it prevent from injuries?

Q.4 a) State the Newton's first law of motion. Give some common examples?

Q.4 b) Enlist some common observations that are caused by the property of inertia?

Q.5 Define Newton's second law of motion.

Q.6 Show the relationship between applied force and the acceleration produced in the body?

Q.7 Enumerate at least three clear differences between mass and weight?

Q.8 a) Define the forces acting on an object in circular motion?

Q.8 b) Draw a figure showing the direction of centripetal force, centrifugal force and velocity of an object along a circular path.

Q.8 c) List down some purposeful uses of centrifuge that human are benefitting everyday.

Q.9 What is force of friction? Explain with two examples from daily life.

Q.10 a) How anti-lock braking system prevents the risk of sliding?

Q.10 b) Enlist any four uses of rolling friction in everyday life?

Q.11 Explore the following phenomenon in relation with dynamics

Q.12 a) When an air filled balloon is released.

Q.12 b) Riding a bicycle needs continuous paddling.

Q.12 c) The biker ridding in the death well.

Q.12 d) You always feel a pullback whenever you pull on your school bag or some heavier object.

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q.1 Find the moment of body of mass 6 kg moving with a velocity of 25 ms^{-1} .

Q.2 What will be the velocity if the momentum becomes 200Ns?

Q.3 A body of mass 10 kg is moving with velocity of 10 ms^{-1} . A force acts for 5 seconds to reduce its velocity to 2 ms^{-1} . Find the momentum of body before and after application of the force on it.

Q.4 (a) A force of 3400 N is applied on a body of mass is 850 kg, find the acceleration produced by the force?

Q.4 (b) How much force should be applied on a body of mass 425kg to produce acceleration same as calculated in part a.

Q.5 Find the mass of a body which is accelerated by applying a force of 200 N, that speeds up it to 36 ms^{-1} .

- Q.6 What should be the acceleration of the same body if the applied force changes to 280N.
- Q.7 An empty car has 1200 kg mass. Its engine can produce acceleration of 4 ms^{-2} . If 300 kg load is added to mass by passengers and luggage. What acceleration the same engine will produce?
- Q.8 The mass of an object is 60 kg, find its weight on
(i) Earth (ii) Moon (iii) Mars assume the acceleration due to gravity on Earth = 9.8 ms^{-2} on Moon = 1.6 ms^{-2} and on Mars = 3.7 ms^{-2}
- Q.9 A car is running on a circular part of highway having about 1000m radius. The mass of car is 600kg and its velocity is 72 kmh. Find
(i) Centripetal force exerted by the car.
(ii) Centripetal acceleration of car.
- Q.10 A block is placed on a wet slippery floor. The mass of block is 15 kg. When it is pulled through a string and spring balance, it shows force equal to 3 N. Find the coefficient of friction. ($F_s = \mu mg$)

Chapter 4: TURNING EFFECT OF FORCES

Section B

2. What do you mean by resolution of a vector? Also split the vector into its perpendicular components.	2025
7. Define torque. Write down its formula and unit. 13. A man is pushing a Trolley for a horizontal ground with a force of 100N making an angle of 60° with ground. Find the horizontal and vertical components of the force.	2024
5. Write three state of equilibrium.	2023
4. Describe three states of equilibrium with one example of each. 9. A mechanic uses a double arm spanner to turn a nut. He applies a force of 15N at each end of the spanner and produces a torque of 60Nm. What is the length of the moment arm of the couple?	2022
7. Find the two rectangular components of a force of 100 N which is acting at an angle of 60° with x-axis. ($\cos 60^\circ = 0.5$, $\sin 60^\circ = 0.866$)	2021
13. A force is acting at an angle 60° with x-axis. If the x-component of the force is 50 Newton. Find resultant force and y-component of the force. ($\sin 60^\circ = 0.866$, $\cos 60^\circ = 0.5$)	2019
7. Write three states of equilibrium. 18. With the help of diagram define negative of a vector and resultant vector.	2017
6. With the help of rectangular components of a vector derive	2016

the equation for the resultant vector $F = \sqrt{F_x^2 + F_y^2}$	
20. Define equilibrium and state two conditions of equilibrium.	
6. With the help of Graphical method add two Vectors $\vec{A}$ and $\vec{B}$.	2015
7. Prove that Moment of the couple is equal to the product of one of the forces and arm of couple.	
8. A 5m long beam pivoted into a wall a force of 100 N is applied vertically at the Far end of beam. Find the Torque produced	2014
Section C	
18. Explain two types of equilibrium with examples on the basis of state of a body. Also write two conditions of equilibrium in detail.	2025
15. Define equilibrium. Explain its types and two conditions.	2024
15. Which process is the reverse of resolution of a force? Define it also prove that	2023
$F = \sqrt{F_x^2 + F_y^2} \text{ and } \theta = \tan^{-1} \frac{F_y}{F_x}$	
18. (b) Define resolution of a vector. Explain the method to resolve a vector into its rectangular components.	2018
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Q.1 Define like and unlike forces.	
Q.2 Describe the head to tail rule of vector addition of forces.	
Q.3 How a force can be resolved in to its perpendicular components? i) Find its horizontal component ii) Find its vertical component	
Q.4 a) How can you determine a force from its rectangular components?	
Q.4 b) Horizontal and vertical components of a force are 4 N and 3 N respectively. Find (i) Resultant force (ii) Direction of resultant	
Q.5 What do you mean by moment of force?	
Q.6 State the principle of moment.	
Q.7 a) Define Center of mass or Center of gravity.	
Q.7 b) How will you determine the Center of mass or Center of gravity?	
Q.8 Define couple as a pair of forces tending to produce torque	
Q.9 State two conditions necessary for an object to be in equilibrium.	
SOME IMPORTANT NUMERICAL FROM NEW BOOK	
Q.1 A pair of like parallel forces 15N each are acting on a body. Find their resultant.	
Q.2 Two unlike parallel forces 10 N each acting along same line. Find their resultant.	

Q.3 Three forces 12 N along x-axis, 8 N making an angle of 45° with x-axis and 8 N along y-axis. i) Find their resultant ii) Find the direction of resultant
Q.4 A gardener is driving a lawnmower with a force of 80 N that makes an angle of 40° with the ground. i) Find its horizontal component ii) Find its vertical component
Q.5 A spanner of 0.3 m length can produce a torque of 300 Nm. i) determine the force applied on it ii) What should be the length of the spanner if torque is to be increased to 500 Nm with same applied force
Q.6 A uniform meter rule is supported at its center is balanced by two forces 12 N and 20 N i) If 20 N force is placed at a distance of 3 m from pivot find the position of 12 N force on the other side of pivot ii) if the 20 N force is moved to 4 cm from pivot then find force to replace 12 N force.
Q.7 A mechanic uses a double arm spanner to turn a nut. He applies a force of 15 N at each end of the spanner and produces a torque of 60 Nm. What is the length of the moment arm of the couple?
Q.8 If he wants to produce a torque of 80 Nm with same spanner then how much force he should apply?
Q.9 A uniform metre rule is balanced at the 30 cm mark when a load of 0.80 N is hung at the zero mark. i) At what point on the rule is the Centre of gravity of the rule? ii) Calculate the weight of the rule

Chapter 5: FORCES AND MATTER

Section B

3. Define pressure. Write down two factors affecting pressure.	2025
4. State and explain Hook's law.	
6. What do you mean by elasticity? State Hooke's Law and write its mathematical expression.	2024
10. Calculate the spring constant which extends by a distance of 3.5 cm when a load of 14 N is hung from its end.	
6. Define pressure on what factors does the pressure depend?	2023
10. A spring has spring constant $K = 30 \text{ Nm}^{-1}$. What load is required to produce an extension of 4 m?	2022
10. Find the pressure exerted by a 450 N woman putting all her weight onto one heel of area $2 \text{ cm} \times 2 \text{ cm}$.	2021
8. Write the name of the law, which states "When a pressure is applied to a liquid it is transmitted equally in all directions."	2018

Describe any one application of this law."	
14. Define pressure. Write its formula and S.I. unit.	2017
12. Explain Hook's Law applied to Helical Spring.	2015

Section C

16. Describe the construction and working of hydraulic machine with the help of diagram.	2023
14. State Pascal's law. Describe construction and working of a hydraulic lift with the help of a diagram.	2021
24. (b) Define Pascal's law and explain any one of its application by diagram.	2016
24. (c) Describe four examples showing surface tension,	2015

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Q.1 State Hooke's law.

Q.2 Define the term pressure.

Q.3 Write down the S.I unit of pressure.

Q.4 Why does pressure increases as you dig deeper; Explain in detail.

a) The pressure exerted by boy's thumb on the head of thumbtack.

b) The pressure of the tip of the thumbtack on the wood.

c) What conclusion can be drawn from answers of part (a) and (b)?

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q.1 Calculate the spring constant for a spring which extends by a distance of 3.5cm when a load of 14N is hung from its end.

Q.2 In the given Table shows the results of an experiment to stretch a spring.

Load (N)	Extension (cm)
0.0	0.0
2.0	80.0
4.0	83.0
6.0	86.0
8.0	89.0
10.0	92.0
12.0	93.0

a) Calculate the spring constant k.

b) Show maximum force at which Hooke's law is applicable.

Q.3 A boy is pressing a thumbtack into a piece of wood with a force of 20 N. The surface area of head of thumbtack is 1cm^2 and the cross-section area of the tip of the thumbtack is 0.01cm^2 . Calculate

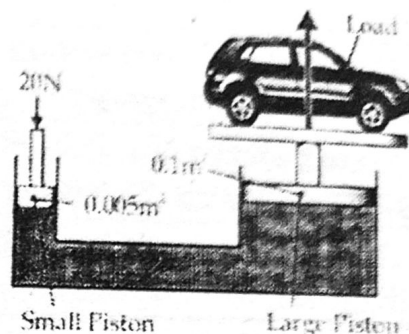
a) The pressure exerted by boy's thumb on the head of thumbtack.

b) The pressure of the tip of the thumbtack on the wood.

c) What conclusion can be drawn from answers of part (a) and (b)?

Q.4 In The Figure shows a basic hydraulic system that has small and large pistons of cross section area of 0.005 m^2 and 0.1 m^2 respectively. A force of 20 N is applied to small piston. Calculate

- The pressure transmitted into hydraulic fluid.
- The force at large piston.
- Discuss the distance travelled by small and large pistons.



Chapter 6:

GRAVITATION

Section B

13. Fatima's weight is 900 N on the Earth's surface. What will be her weight at the surface of moon?	2025
11. A planet has mass four times of earth and radius two time that of earth. If the value of "g" on the surface of earth is 10 m/s^2 What will be its acceleration due to gravity on the planet?	2024
7. Write down any three characteristics of the Gravitational force.	2023
12. If weight of Amna is 200 N at the surface of the earth. Find mass of amna.	
12. Calculate the orbital velocity of artificial satellite required moving around the earth if radius of earth is $6 \times 10^6 \text{ m}$ and the value of "g" is 10 m/s^2	2019
4. Compute the gravitational force of attraction between two boys of masses 50 kg and 40 kg respectively apart from each other by 2 m .	2016
22. Define orbital velocity and derive the formula $V = \sqrt{\frac{GM_e}{R}}$	2015
3. Write the approximate value of mass of our galaxy, earth and moon,	
8. With the help of two bodies mathematically express the law of universal gravitation in three steps.	

Section C

14. State Newton's law of gravitation. How can this law be used to derive the formula for calculating the mass of earth?	2025
17. Calculate the mass of earth by using Newton's law of	2024

gravitation.	
20. (b) State the Law of Universal Gravitation. Prove that $M_e = \frac{gR_e}{G}$	2018
23. (b) State Newton's Law of gravitation and derive the formula for mass of earth with its help.	2016

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Q.1 a) Why we do not feel the gravitational force of attraction from the objects around us?

Q.1 b) Define Gravitational field with an example.

Q.2 a) Write down any three characteristics of Gravitational force?

Q.2 b) Define gravitational field strength.

Q.3 a) State & explain Newton's law of gravitation?

Q.3 b) Define 'field force'.

Q.4 a) Why weight of an object does not remain same every where on Earth?

Q.4 b) Why the unit of weight is Newton? Explain.

Q.5 Define weight and write down its equation?

Q.6 a) Your weight decreases as you go up at high altitudes, without dieting. Explain.

Q.6 b) If you step on a scale and it gives reading 55kg, is that a measure of your weight. If not then which physical quantity it shows?

Q.7 Calculate the mass of Earth by using Newton's law of gravitation.

Q.8 If " M_E " is the mass of Earth, " R_E " radius of Earth, " G " is universal gravitational constant, then find acceleration due to gravity " g ";

i) On the surface of Earth. ii) At the centre of Earth.

Q.9 Evaluate the acceleration due to gravity in terms of mass of Earth " M_E ", radius of Earth " R_E " and universal gravitational constant " G ";

i) At a distance, twice the Earth's radius.

ii) At a distance, one half the Earth's radius.

Q.10 Write the name of any one natural satellite.

Q.11 a) Write down the names of four different types of orbit.

Q.11 b) Define the terms

i) Critical Velocity. ii) Communication Satellite.

Q.12 Derive the expression for the motion of a satellite. $v = \sqrt{\frac{GM}{R+h}}$

Q.13 a) Differentiate between the natural satellite and artificial satellite.

Q.13 b) Name the parameters on the basis of which orbits are characterized.

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q.1 Determine the gravitational force of attraction between Urwa and

Ayesha standing at a distance of 50m apart. The mass of Urwa is 60kg and that of Ayesha is 70kg.

Q.2 Weight of Rani is 450N at the surface of Earth. Find her mass?

Q.3 Weight of Naveera is 700N on the Earth's surface. What will be Naveera's weight at the surface of Moon?

Q.4 A planet has mass four times of Earth and radius two times that of Earth. If the value of "g" on the surface of Earth is 10ms^{-2} . Calculate acceleration due to gravity on the planet.

Q.5 Calculate the speed of a satellite which orbits the Earth at an altitude of 400 kilometers above Earth's surface.

Chapter 7: PROPERTIES OF MATTER

Section B

9. A cylinder contains 120cm^3 of air at pressure of 140kPa. What will be its volume if the pressure on it is increase to 840kPa?	2025
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8. A cylinder contains 60cm^3 of air at a pressure of 140kpa. What will its volume be if the pressure on it is increased to 200kPa?	2023
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11. A cylinder contains 75cm^3 of air at a pressure of 120K Pa. What will be its volume if the pressure on it, is increased to 360 KPa? While its temperature remains constant.	2022
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Section C

17. Which variables describe completely the behavior of a gas on the basis of kinetic molecular theory (KMT)? Explain any one of them.	2025
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16. State and explain Boyle's Law. Write its one application.	2024
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17. Which variables describe completely behavior of a gas on the basic of Kinetic Molecular Theory (KMT)? Explain any one of them.	2023
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SOME IMPORTANT QUESTIONS FROM NEW BOOK

Q.1 a) "The particles are free to move within the material, has a fixed volume but takes up the shape of its container", which state of matter is being described here?

Q.1 b) Write a similar description of the particles that make up a solid.

Q.1 c) Write down any two properties of a solid.

Q.2 a) Why the kinetic model of matter is called kinetic?

Q.2 b) In which state of matter the molecules are widely separated?

Q.2 c) In which state of matter the molecules are most closely packed?

Q.2 d) In which state of matter molecules can move freely at high speed?

Q.3 a) By using kinetic molecular theory explain why we can walk through air, swim through water but can not walk through a solid wall.

Q.3 b) In which state of matter do the molecules have minimum kinetic energy?

Q.3 c) Which state of matter is highly incompressible?

Q.4 A sample of a gas is in a sealed test tube is cooled. Describe what happens to:

- The size of the molecules.
- The speed at which molecules move.
- The number of the molecules.
- The pressure inside the tube.
- The state of the gas.

Q.5 An inflated car tyre is considered to have a constant volume, regardless of any changes in temperature or pressure. Use the kinetic molecular theory of gases to answer the following:

- How does air in the tyre exert a pressure on the walls of the tyre?
- Why is the pressure the same at all points on the inside wall of the tyre?
- More air is pumped into the tyre while the temperature is kept constant until there are twice as many molecules as before. Explain why you would expect the pressure to be doubled.

Q.6 Describe the following:

- What happens to the motion of the molecules of a gas when it cools down?
- What happens to the motion of a liquid when it cools down?

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q.1 The pressure on 9cm^3 of oxygen gas is doubled at a fixed temperature. What will its volume become?

Q.2 A container holds 30m^3 of air at a pressure of 150000Pa . If the volume changed to 10m^3 by decreasing load on the piston. What will the pressure of the gas become? Assume that its temperature remains constant.

Q.3 Air at atmospheric pressure of 760 mm of Hg is trapped inside a container available with a moveable piston. When the piston is pulled out slowly so that the volume is increased from 100dm^3 to 150dm^3 , the temperature remaining constant. What will be the pressure of the air becomes?

Q.1 The pressure on 9cm^3 of oxygen gas is doubled at a fixed temperature. What will its volume become?

Chapter 8: ENERGY SOURCES AND TRANSFER OF ENERGY

Section B	
6. What is wind energy? Write down its two applications. 10. Calculate the power of a machine. If the machine performs 1800 Joules of work in 30 minutes.	2025
5. What is potential energy? Derive $P.E = mgh$ 12. What will be the kinetic energy of a boy of mass 50kg driving a bike with the velocity of 2 ms^{-1}	2024
9. If the efficiency of a machine is 50% and its output is 100J then calculate its input.	2023
6. Write down three applications of wind energy. 12. Calculate the power of a machine. If the machine performs 900 Joules of work in 30 minutes.	2022
13. Define Kinetic energy and derive its equation. $K.E = \frac{1}{2} mv^2$	2021
16. What is the Kinetic Energy (K.E.) of 2000 Kg car travelling at a velocity of 90 km/h?	2017
9. Define Power and derive the equation $P = FV$ 13. With what constant velocity can a 1960 watt motor raise a mass of 100 kg?	2016
10. Describe inter conversion of kinetic and potential energy.	2015
Section C	
18. State law of conservation of energy. Also explain any five renewable energy sources.	2023
16. Define Kinetic energy and Potential energy and derive equation for gravitational potential energy $P.E = mgh$.	2022
20. (a) Define Potential and Kinetic Energy, Also derive their equations: $P.E = mgh$ and $K.E = \frac{1}{2} mv^2$	2019
18. (a) Define K.E, and P.E. Derive the equation $K.E = \frac{1}{2} mv^2$	2018
SOME IMPORTANT QUESTIONS FROM NEW BOOK	
Q.1 a) Define work?	
Q.1 b) Derive the equation; work = $Fd \cos\theta$.	
Q.2 a) Define Kinetic energy	b) Derive the equation.
Q.3 a) Define Potential Energy	b) Derive the equation. $PE = mgh$
Q.4 a) Why fossil fuel energy is called nonrenewable source?	
Q.4 b) Define solar energy and its importance in Pakistan?	
Q.5 Write notes on Tidal energy and Geothermal energy.	
Q.6 a) What is wind energy?	
Q.6 b) Write any three applications of wind energy?	
Q.7 a) Write the name of any one radioactive element which is used as	

source of nuclear energy.

Q.7 b) Write the names of any one device that can convert solar energy into heat energy.

Q.7 c) Write the names of any two devices that can convert solar energy into electrical energy.

Q.8 Write a note on renewable energy sources?

Q.9 Write a note on non-renewable energy sources?

Q.10 What is the difference between renewable of nonrenewable energy sources?

Q.11 Make a table of renewable and non-renewable energy sources from the following: Uranium, Solar, Coal, Wind, Natural gas, Tidal, Biomass, Hydroelectricity.

Q.12 Write a note on efficiency.

Q.13 Define power.

Q.14 Define Watt.

Q.15 Write down the names of any two larger units of power.

SOME IMPORTANT NUMERICAL FROM NEW BOOK

Q.1 How much work is needed to move horizontally a body 20m by a force of 30N, the angle between the body and the horizontal surface is 60° ?

Q.2 How much work is done, if a crate is moved at a distance of 50m, when a force of 30N is applied along the surface.

Q.3 What is the work done by Usman? If a bar of weight 100 N is brought by him from A to B, then brought back to A.

Q.4 What will be the Kinetic energy of a boy of mass 50kg driving a bike with velocity of 2ms^{-1} .

Q.5 a) If LED screen of mass 10kg is lifted up and kept it on a cupboard of height 2m. Calculate the potential energy stored in the LED screen.

Q.6 Calculate the potential energy of 3kg water raised to the tank at the roof of a home 4m high. (Assume $g=10\text{ms}^{-2}$)

Q.7 Calculate the efficiency of a machine which consumes 200 J of energy and performs 50J of work.

Q.8 If the efficiency of a machine is 70% and its output is 100 J then calculate its input.

Q.9 Which machine is more efficient, machine "A" which has an output of 200J after consuming 400J of energy or machine "B" which has an output of 300J after consuming 450J of energy?

Q.10 The energy of 600J dissipated by a bulb in 50 minutes. Find the power of the bulb.

Q.11 a) Convert 20watt into horse power.

Q.11 b) Calculate the power of a machine, if it does 40 Joules of work in 10 sec.

Q.12 A student of weight 400N takes 5 sec to climb up an obstacle of height 2m. Calculate the power consumed?

Q.13 If a machine consumes 250J of energy per hour then what will be its power?

Chapter 9: THERMAL PROPERTIES OF MATTER

Section B

12. How much heat energy will be required to raise the temperature of 50g of water from 40°C to 70°C ? Where is Specific heat of water is $4200\text{J/kg}^{-1}\text{K}^{-1}$	2025
4. Write down three difference between Boiling Point and Evaporation.	2024
10. 2Kg of copper requires 2050J of heat to raise its temperature through 10°C . Calculate the heat capacity of the sample.	2023
13. How much heat energy will be required to raise the temperature of 50 g of water from 40°C to 70°C ? Specific heat of water is $4200\text{J kg}^{-1}\text{K}^{-1}$	2022
14. Find the amount of heat required to raise the temperature of 100 gm of water from 10°C to 60°C . (Sp. Heat of water= $4200\text{J/Kg}^{\circ}\text{C}$)	2019
2. Write down three differences between heat and temperature. 9. A steel rod has a length of 10 meters at a temperature of 25°C . Calculate the increase in length if it is heated to 35°C . 15. What is Bimetallic Strip? Write the names of its two applications	2017
13. A steel rod has a length of 10m at a temperature of 25°C calculate the increase in length if it is heated to 35°C	2015

Section C

19. Define the following: (i) Specific Heat capacity (ii) Latent heat of vaporization (iii) Latent heat of fusion Also write three benefits of high specific heat capacity of water.	2025
18. Define Thermal expansion. Derive $\Delta L = \alpha L \Delta T$	2024
19. Define evaporation. Also write down factors affecting surface evaporation process.	2023
15. Define linear expansion and volumetric expansion and derive the equation $\Delta L = \alpha L \Delta T$	2022
15. Define co-efficient of linear expansion and co-efficient of volumetric expansion. Derive the equation: $L_2 = L_1(1 + \alpha \Delta T)$	2021
19. (a) Define co-efficient of linear expansion and derive the relation $L_2 = L_1[1 + \alpha \Delta T]$	2018

23.(a) Derive the equation; $\beta = 3\alpha$

2016

SOME IMPORTANT QUESTIONS FROM NEW BOOK

Q.1 a) Define Heat and write its SI unit.

Q.1 b) Why does heat flows from hot body to cold body?

Q.2 a) Explain three different scales of temperature along with their main uses.

Q.2 b) Differentiate between heat and temperature.

Q.3 a) Explain specific heat capacity.

Q.3 b) How would you find the specific heat of a solid?

Q.4 Explain the effects of large specific heat of water with examples from our daily life.

Q.5 Define heat of fusion with the help of an experiment.

Q.6 Differentiate between heat of fusion and heat of vaporization

Q.7 Demonstrate heat of fusion and heat of vaporization by the help of heating ice graph.

Q.8 Explain in detail, why evaporation causes cooling?

Q.9 Differentiate between evaporation and boiling.

Q.10 Write any four factors that influence the surface evaporation.

Q.11 Write down the freezing and boiling points of following

i) Acetic acid ii) Benzene iii) Chloroform iv) Water

Q.12 Why solids increases in size on heating? Explain.

Q.13 a) Draw the diagram, showing real and apparent expansion of liquid. Label the diagram properly.

Q.13 b) Why small gaps are left at the joints of sections of railway tracks? Explain the phenomenon involved in it.

SOME IMPORTANT NUMERICAL FROM NEW BOOKQ.1 Convert 30°C into Kelvin and Fahrenheit Scale.Q.2 Convert 212°F into Celsius and Kelvin.Q.3 How much heat is required to boil 3 kg water which is initially 10°C ?Q.4 2kg of copper requires 2050J of heat to raise its temperature through 10°C . Calculate the heat capacity of the sample.Q.5 An iron block of volume 3m^3 is heated, so that its temperature changes from 25°C to 100°C . If the coefficient of linear expansion of iron is $11 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$. What will be the new volume of the iron block after heating?**MISCELLANEOUS QUESTIONS****DEFINE / DEFINITIONS****QUESTIONS****CHAPTER****YEAR**